

Normalization Report - Hotel Reservation System

Objective

Design a scalable, normalized SQL Server database for customer management, room booking, availability tracking, billing, payment processing, room category pricing, audit logging, and hotel reporting.

1NF

Repeating room and service details are separated into ReservationRoom and ReservationService. Customer, room, category, service, and payment data are stored as atomic values in separate tables.

2NF

Partial dependency is removed. Room attributes depend only on RoomID, category pricing depends only on CategoryID, service price depends only on ServiceID, and bridge-table attributes depend on the full relationship.

3NF

Transitive dependency is removed. Customer contact data, staff data, room category prices, payment methods, and service prices are maintained only in their own master tables. Billing totals are calculated through SQL logic instead of duplicated.

BCNF

Every determinant is a candidate key or primary key. Unique attributes such as customer email, phone, identity proof number, room number per hotel, category name, service name, and payment method name are constrained.

Entity Relationship Justification

Relationship	Cardinality	Justification
Hotel - Room	1:M	One hotel owns multiple rooms.
RoomCategory - Room	1:M	One category classifies multiple rooms.
Customer - Reservation	1:M	One customer can make many reservations.
Reservation - Room	M:M	Resolved using ReservationRoom to allow multi-room bookings.
Reservation - Service	M:M	Resolved using ReservationService for chargeable services.
Reservation - Payment	1:M	A reservation may have multiple payments.

Integrity and Validation

The schema uses primary keys, foreign keys, NOT NULL, UNIQUE, CHECK constraints, defaults, triggers for double-booking prevention, and audit logging for insert, update, and delete activity.

Reporting Support

Views, indexes, functions, joins, subqueries, and aggregation queries support availability reports, occupancy rate, customer history, revenue summaries, pending balances, service usage, and audit review.